

POD PHOENIX

SQL CAPSTONE PROJECT

07/03/26

Business overview

- **NextGen Corp.** is a growing technology company operating in the software and hardware space. As the company expands, HR needs a data-driven understanding of employee retention, performance, and salary patterns. Increasing concerns around employee turnover, performance variability, and salary fairness make it important to identify trends and provide recommendations that support workforce stability and organizational success.



The background is a blurred image of a document. It features a line graph with a jagged line and a dotted trend line. A pen is visible in the upper right corner, pointing towards the graph. Some numbers are visible on the graph, including '2,5' and '2,47'.

EMPLOYEE ANALYSIS

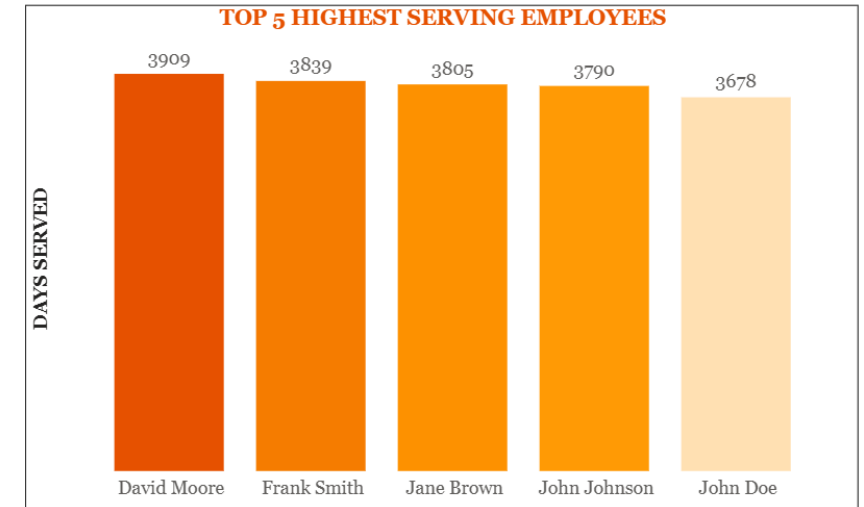
Who are the top 5 highest serving employees?

Insight

- The analysis identifies the employees with the longest tenure in the organization. These long-serving employees contribute valuable institutional knowledge, experience, and stability to the workforce.

Recommendation

- NextGen Corp. should recognize and reward long-serving employees through career advancement opportunities, leadership roles, and retention incentives to maintain workforce stability.



employee_id [PK] integer	full_name text	hire_date date	tenure interval
8	David Moore	2015-06-30	10 years 8 mons 15 days
48	Frank Smith	2015-09-08	10 years 6 mons 7 days
67	Jane Brown	2015-10-12	10 years 5 mons 3 days
44	John Johns...	2015-10-27	10 years 4 mons 19 days
57	John Doe	2016-02-16	10 years 28 days

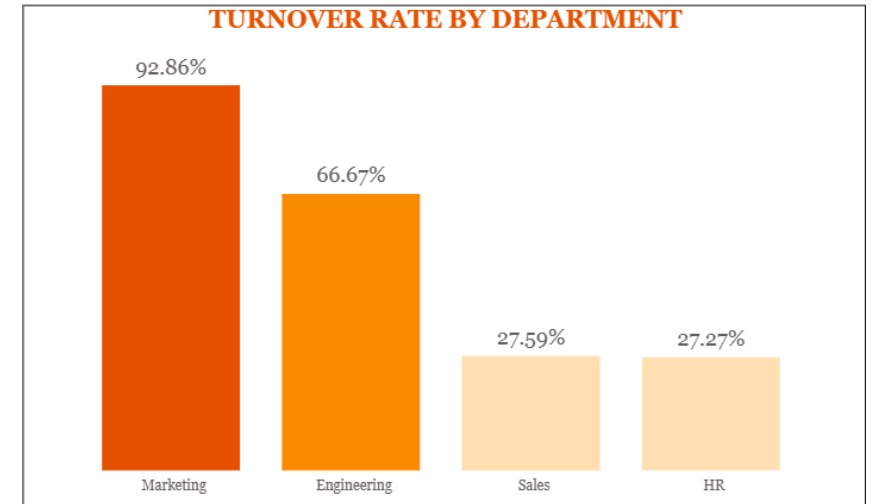
What is the turnover rate for each department?

Insight

- Turnover rates vary across departments, indicating that some teams may experience greater retention challenges. Higher turnover in certain departments may reflect workload pressures, leadership issues, or limited career development opportunities.

Recommendation

- Management should investigate departments with higher turnover rates and implement targeted retention strategies such as improved work conditions, mentorship programs, and stronger leadership support.



department_id [PK] integer	department_name character varying (30)	turnover_rate numeric
3	Marketing	92.86
1	Engineering	66.67
2	Sales	27.59
4	HR	27.27

Which employees are at risk of leaving based on their performance?

Insight

- Employees with lower performance scores were classified as **high risk**, suggesting a potential link between low performance and employee disengagement. Identifying these employees early helps the organization address potential turnover risks.

Recommendation

- HR should introduce performance improvement plans, training programs, and regular feedback sessions to support employees with lower performance scores and reduce potential turnover.

RISK OF LEAVING BY PERFORMANCE			
Employee id	Fullname	Performance score	Risk
22	Grace Wilson	3.43	High Risk
1	Jane Wilson	3.58	Medium Risk
12	Eve Davis	3.65	Medium Risk
97	David Smith	3.72	Medium Risk
16	Charlie Lee	3.77	Medium Risk
40	Alice Wilson	3.77	Medium Risk

employee_id	fullname	performance_score	employee_status
[PK] integer	text	numeric	text
22	Grace Wilson	3.43	High Risk
1	Jane Wilson	3.58	Medium Risk
12	Eve Davis	3.65	Medium Risk
97	David Smith	3.72	Medium Risk
40	Alice Wilson	3.77	Medium Risk

What are the main reasons employees are leaving the company?

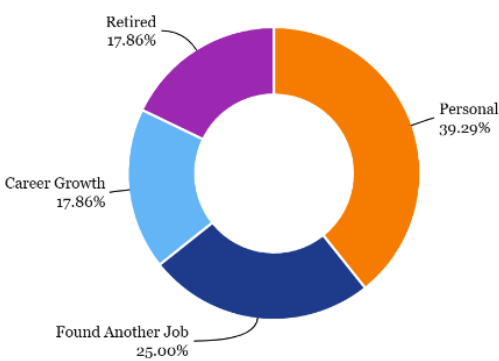
Insight

- The analysis shows that employees leave the company mainly due to **personal reasons, better job opportunities, career growth, and retirement**. While retirement is expected, other reasons may indicate gaps in employee satisfaction and career development.

Recommendation

- NextGen Corp. should strengthen career development pathways, review employee engagement initiatives, and ensure competitive compensation to reduce voluntary employee turnover.

REASONS EMPLOYEES LEAVE THE COMPANY



total_employee bigint	reason_for_leaving text
11	Personal
7	Found Another Job
5	Career Growth
5	Retired

Performance Analysis



How many employees has left the company?

Insight

The number of employees leaving the organization provides an overall measure of workforce stability. Monitoring this metric helps HR identify whether turnover levels remain within acceptable limits.

Recommendation

NextGen Corp. should track employee turnover trends regularly and implement proactive retention strategies to maintain a stable workforce.

28

Employees Left

total_employee_left
bigint



28

How many employees have a performance score of 5.0 / below 3.5?

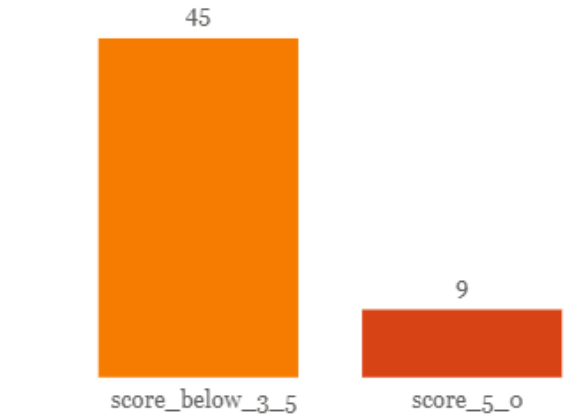
Insight

- Employees with a performance score of **5.0 represent top performers**, while those scoring **below 3.5 may require additional support to improve productivity**. This distribution highlights the variation in employee performance across the company.

Recommendation

- The company should focus on retaining top-performing employees through recognition and incentives, while providing training and development opportunities for lower-performing employees.

PERFORMANCE SCORE BY EMPLOYEES



category	employee_count
text	bigint
score_5_0	9
score_below_3_5	45

Which department has the most employees with a performance of 5.0 / below 3.5?

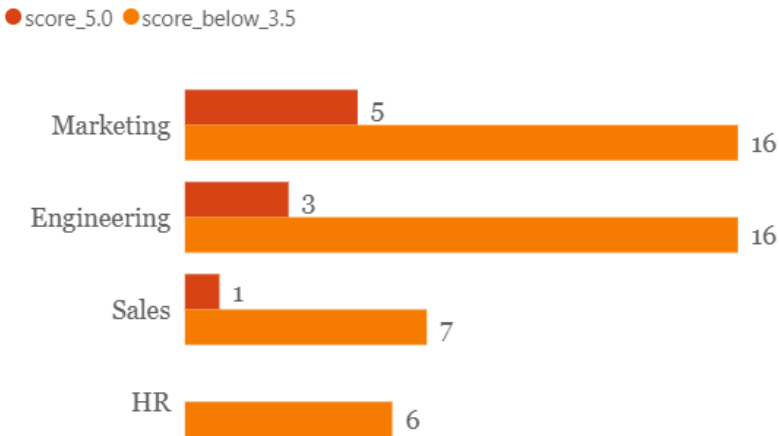
Insight

- Some departments have more high-performing employees while others have a higher concentration of low-performing employees. This difference may reflect variations in leadership quality, team collaboration, or available resources.

Recommendation

- NextGen Corp. should evaluate practices in high-performing departments and replicate successful strategies across other teams to improve overall performance.

PERFORMANCE SCORE BY DEPARTMENT



department_name character varying (30)	category text	employee_count bigint
Marketing	score_below_3_5	16
Engineering	score_below_3_5	16
Sales	score_below_3_5	7
HR	score_below_3_5	6
Marketing	score_5_0	5
Engineering	score_5_0	3
Sales	score_5_0	1

What is the average performance score by department?

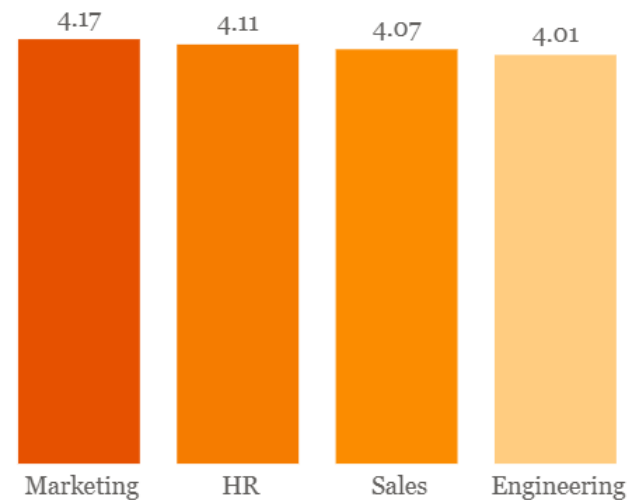
- Insight

- The average performance score by department highlights differences in productivity across teams. Departments with higher average scores demonstrate stronger employee performance and operational efficiency.

- Recommendation

- Management should analyze the factors contributing to higher performance in certain departments and apply these best practices to departments with lower average performance scores.

AVERAGE PERFORMANCE BY DEPARTMENT



department_id [PK] integer	department_name character varying (30)	avg_performance_score numeric
3	Marketing	4.13
1	Engineering	4.10
4	HR	4.05
2	Sales	4.00

SALARY ANALYSIS



What is the total salary expense for the company?

Insight

The total salary expense represents the organization's overall investment in employee compensation. Understanding this cost is essential for budgeting and evaluating workforce sustainability.

Recommendation

- NextGen Corp. should continuously monitor salary expenses and ensure compensation aligns with employee productivity and organizational growth.

£4.850M

Total Salary Expenses

total_salary_expense
text



£4,850,000.00

What is the average salary by job title?

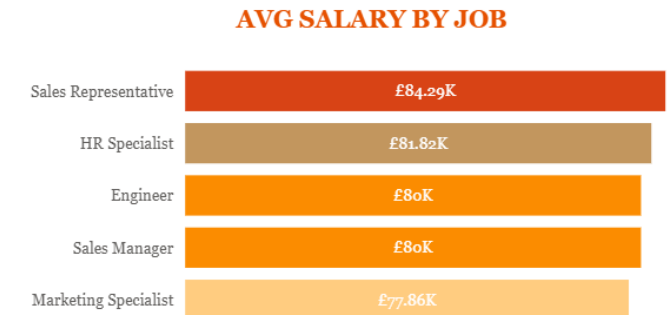
Insight

- Average salary varies across job titles, reflecting differences in responsibility levels, required skills, and seniority. Higher salaries are typically associated with leadership or specialized roles.

Recommendation

- The company should ensure salary structures remain competitive and aligned with industry standards to attract and retain skilled employees.

job_title	avg_salary
character varying (30)	text
Sales Representative	£84,285.71
HR Specialist	£81,818.18
Engineer	£80,000.00
Sales Manager	£80,000.00
Marketing Specialist	£77,857.14



How many employees earn above 80,000?

- **Insight**

- Employees earning above 80,000 likely occupy senior or specialized positions that require advanced expertise. These employees represent critical talent within the organization.

- **Recommendation**

- NextGen Corp. should focus on retaining high-value employees through career development opportunities, leadership pathways, and competitive benefits.

26

Employees Above 80000

```
total_employees_above_80000  
bigint
```



26

How does performance correlate with salary across departments?

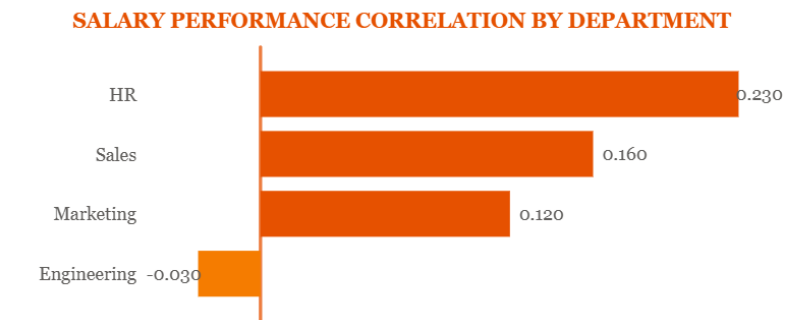
Insight

- The correlation between salary and performance helps determine whether employee compensation reflects productivity. A stronger positive correlation suggests that higher-performing employees receive higher pay.

Recommendation

- NextGen Corp. should ensure salary structures fairly reward employee performance, which can increase motivation, productivity, and employee satisfaction.

department_name character varying (30)	salary_perf_corr numeric
HR	0.23
Sales	0.16
Marketing	0.12
Engineering	-0.03



Dashboard Overview

- To further explore the insights generated from the SQL analysis, an interactive Power BI dashboard was created. The dashboard visualizes key metrics related to employee retention, performance trends, and salary distribution across departments.

NEXTGEN CORP DATA
ANALYSIS

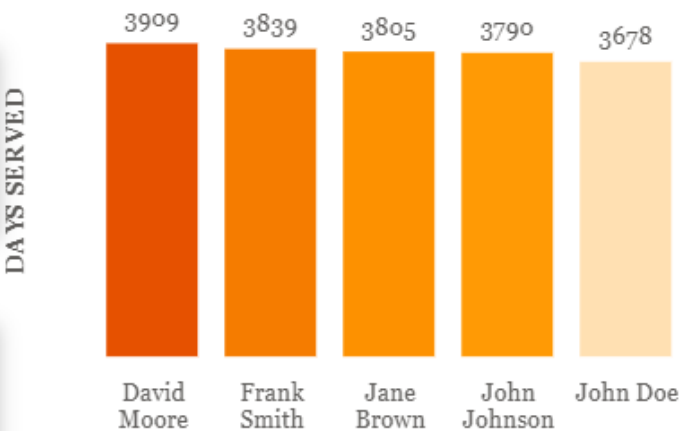
60
Total Employees

28
Employees Left

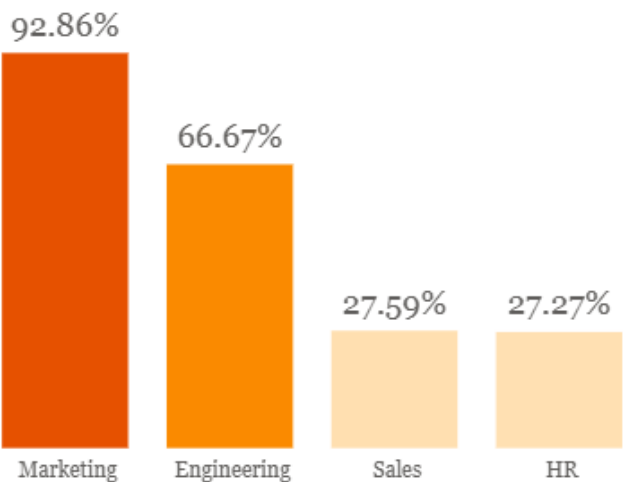
46.67%
Turnover Rate %

4.09
Average Performance

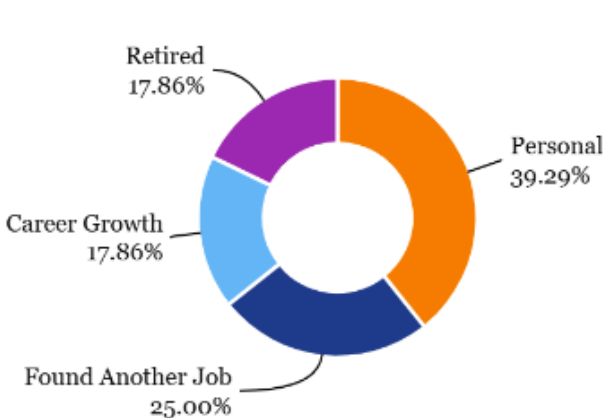
TOP 5 HIGHEST SERVING
EMPLOYEES



TURNOVER RATE BY DEPARTMENT



REASONS EMPLOYEES LEAVE THE
COMPANY



TOP 5 HIGHEST SERVING
EMPLOYEES

Full Name	Tenure
John Doe	10 years 25 days
John Johnson	10 years 4 mons 16 days
Jane Brown	10 years 5 mons
Frank Smith	10 years 6 mons 4 days
David Moore	10 years 8 mons 12 days

RISK OF LEAVING BY PERFORMANCE

Employee id	Fullname	Performance score	Risk
22	Grace Wilson	3.43	High Risk
1	Jane Wilson	3.58	Medium Risk
12	Eve Davis	3.65	Medium Risk
97	David Smith	3.72	Medium Risk
16	Charlie Lee	3.77	Medium Risk
40	Alice Wilson	3.77	Medium Risk

- ☐ High Risk
- ☐ Low Risk
- ☐ Medium Risk
- ☐ (Blank)
- ☐ Engineering
- ☐ HR
- ☐ Marketing
- ☐ Sales

60
Total Employees

4.09
Average Performance

5M
Total Salary Expenses

80.83K
Average Salary

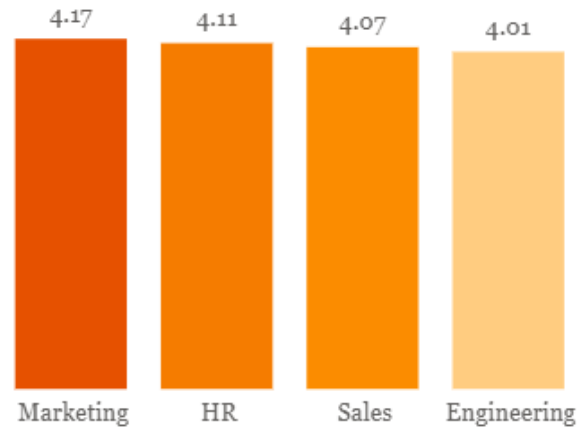
26
Employees Above 80000

- ☐ score_5.0
- ☐ score_below_3.5

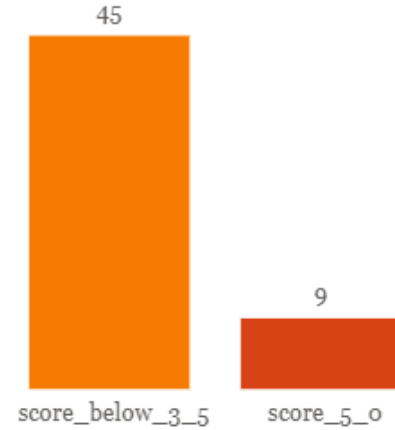
Job Title
All

Departments
All

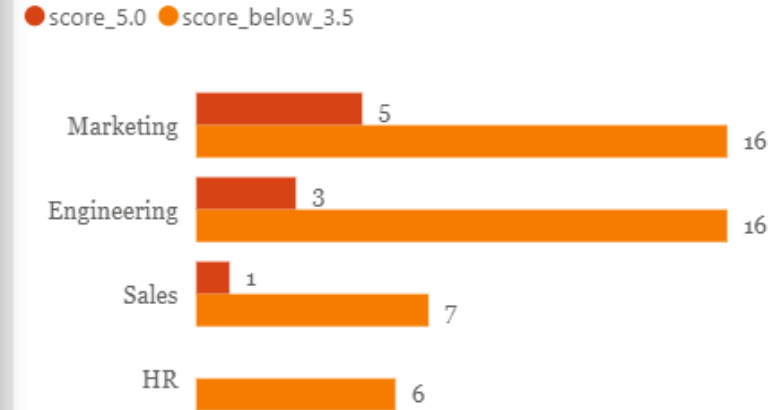
AVERAGE PERFORMANCE BY DEPARTMENT



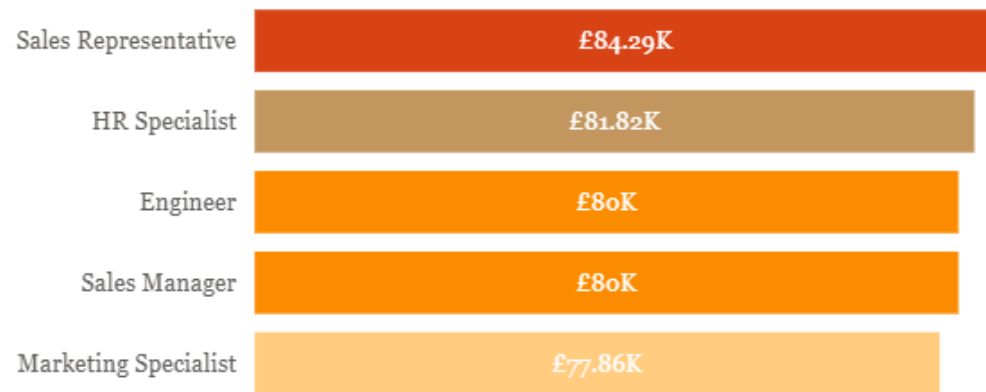
PERFORMANCE SCORE BY EMPLOYESS



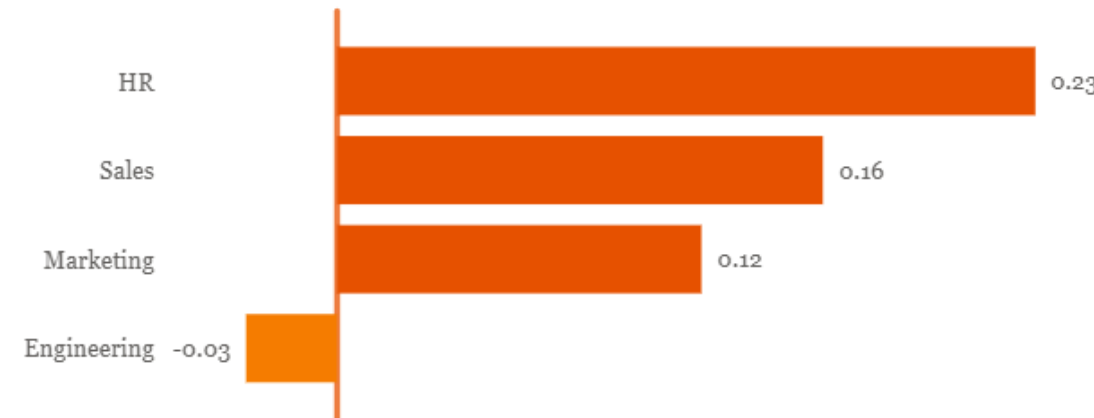
PERFORMANCE SCORE BY DEPARTMENT

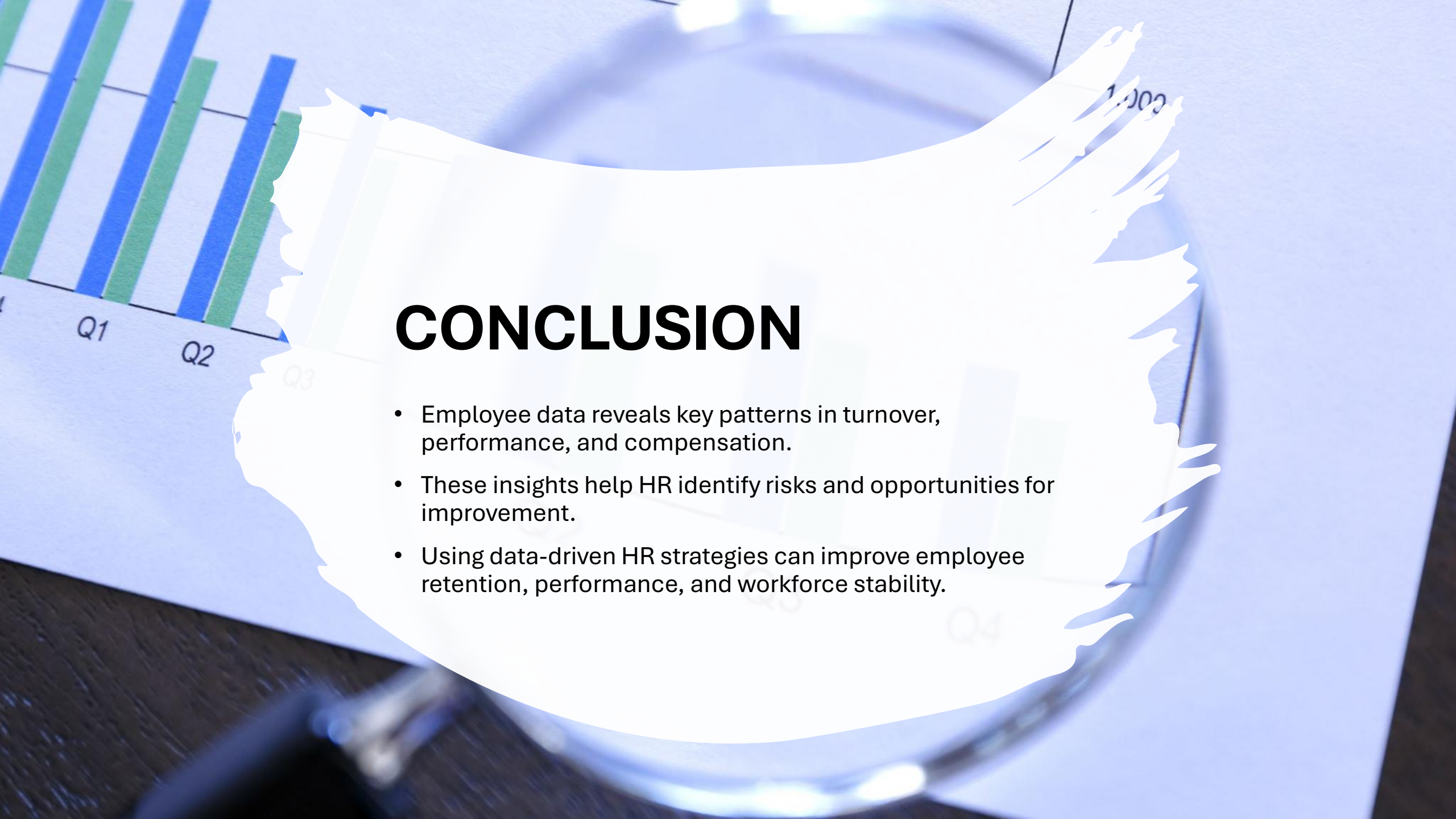


AVG SALARY BY JOB



SALARY PERFORMANCE CORRELATION BY DEPARTMENT





CONCLUSION

- Employee data reveals key patterns in turnover, performance, and compensation.
- These insights help HR identify risks and opportunities for improvement.
- Using data-driven HR strategies can improve employee retention, performance, and workforce stability.



THANK YOU

